

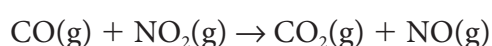
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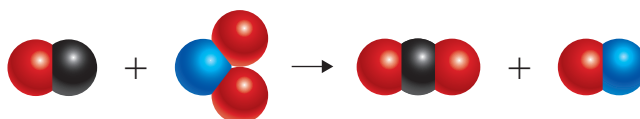
ACTIVITY SHEET

9.3 Modelling successful and unsuccessful collisions

When carbon monoxide and nitrogen dioxide molecules collide, they sometime react to form carbon dioxide and nitric oxide according to the equation:



The reaction can be modelled using space-filling drawings of molecules.



Successful collisions depend on the molecules having a combined kinetic energy equal to or greater than the activation energy for the reaction, and the collision occurring at the right orientation.

Instructions

Using the colour coding shown above, draw space-filling molecular model diagrams to illustrate the following types of collisions. Show the direction and speed of molecules using arrows.

	Diagram	Result
Sufficient energy at the correct orientation		
Correct orientation but insufficient energy		

	Diagram	Result
Sufficient energy but wrong orientation		
Insufficient energy and wrong orientation		